

YELDEN PROTOCOL

Accountability for Autonomous Trading Agents

“An agent that cannot lose cannot be trusted.” — Taleb, Skin in the Game

Version 16.0 · October 2026 · Polygon Mainnet (Chain ID 137)

AI-AgentRegistry

0xC6Aef54A0ea6AbFcC9fe61154b9b357E0DDF113E

verify: polygonscan.com/address/0xC6Aef54A0ea6AbFcC9fe61154b9b357E0DDF113E

yelden.fund

00 — Executive Summary

Autonomous trading agents manage capital at scale. None are accountable.

An agent that underperforms today can simply stop, restart under a new address, and continue operating with no track record, no consequence, and no mechanism for capital allocators to distinguish skill from luck. This is not a theoretical problem — it is the current state of the entire agent economy.

Yelden Protocol is the accountability layer for autonomous trading agents. We provide:

- YAAF — a multi-component, literature-grounded score validated empirically to predict future returns.
- AI-Agent-Registry — an on-chain registry on Polygon that stores scores, enforces fees, and slashes underperformers.
- A fee model in USDC/USDT/MATIC where excellence is free and mediocrity costs.

The core empirical result: YAAF's S_RAW score demonstrated an Information Coefficient of $\rho=0.4362$ ($p=0.0005$, $n=60$) against future returns on GMX V2 with real R-multiples. Sharpe ratio on the same sample achieved only $\rho=0.1747$ ($p=0.18$). YAAF is 2.5x more predictive than Sharpe.

Current state: 4 live trading agents in production (Pepperstone, FTMO, OKX, IBKR), 478 traders scored in the observatory across 3 venues, AI-Agent-Registry deployed on Polygon mainnet.

01 — The Problem

Three structural failures define the current agent economy.

1.1 Agents manage capital. None are accountable.

Autonomous trading agents execute real capital across DeFi, FX, and equities. They are faster than humans, run continuously, and operate without oversight. But there is no infrastructure that holds them accountable when they fail.

1.2 Track records do not survive.

An agent that loses money can terminate and restart under a new address. Its history is erased. No allocator can verify performance over time. There is no persistent, verifiable, cross-venue track record.

1.3 No common scoring standard.

Every protocol has its own metrics. Sharpe here, ROI there, win rate elsewhere. There is no canonical score that is grounded in the quantitative finance literature, scale-free, anti-overfit, anti-gaming, and empirically validated against future returns.

02 — What Yelden Solves

Yelden introduces an accountability-first framework where every economic flow is conditioned on verifiable performance.

Problem	Yelden's solution
Agents without consequence	slash() on verified drawdown / capital loss
No persistent track record	SISTEMA score compounds over rounds via EMA
No common standard	YAAF — 11 components, literature-grounded
No anti-gaming	Volume Axiom — hard gate on real position size
No empirical validation	IC test: $\rho=0.4362$, $p=0.0005$ (GMX V2, $n=60$)

03 — The YAAF Score

3.1 S_RAW v5 — 11 Components + 2 Penalties

S_RAW is computed from 11 normalised performance metrics. The weights below are the calibrated values of the Formal Specification; they sum to 0.96 and are divided by that sum before being applied, so that S_RAW is bounded in [0, 100] while relative importance is preserved exactly (Amendment v5.0.1). CVaR and Drawdown enter as direct subtractions from the weighted sum.

#	Component	Weight	Formula
1	Sharpe × DSR	0.14	$\min(\text{SR}/2.5, 1) \times 100 \times \text{DSR}$
2	Sortino × CF	0.07	$\min(\text{Sortino}/8, 1) \times 100 \times \min(\text{n_loss}/20, 1)$
3	Win Rate	0.08	$\text{WR} \times 100$
4	Profit Factor	0.10	$\min((\text{PF}-1)/1.5, 1) \times 100$
5	Avg R	0.07	$\min(\text{AvgR}/1.5, 1) \times 100$
6	Expectancy	0.09	$\min(\max(\text{E}(\text{R}), 0)/0.6, 1) \times 100$
7	Inv. Volatility	0.05	$\max(1 - \sigma_d/2, 0) \times 100$
8	Stability ²	0.09	$(1/(1+\sigma_{\text{mSharpe}}))^2 \times 100$
9	Smoothness ²	0.08	$\min(\text{Calmar}/4, 1)^2 \times 100$
10	Pf Percentile	0.12	% of 1,000 random portfolios beaten
11	Mc (Independence)	0.07	$(1 - \beta)^2 \times 100$
—	CVaR penalty	—	$\min(\text{CVaR}/2, 1) \times 15$
—	Max DD penalty	—	$\text{MaxDD}\% \times 0.20$

Component Specifications

Four of the components above admit more than one reading. The Formal Specification fixes them as follows, so that any third party holding the same trade record can reproduce a published score.

Pf Percentile (c10). $K = 1,000$ vectors are drawn from $N(0, \sigma_r)$ with the same length as the agent's R-multiple series. The component is the percentage of those random portfolios whose annualised Sharpe the agent's real Sharpe exceeds. $\text{Pf}\% = 97$ means the strategy beats 97% of random portfolios with identical risk and sample size.

Mc — Independence (c11). β is the correlation between the agent's daily aggregated R and the daily returns of the underlying market. The quadratic form rewards edge that is independent of market direction, and penalises a track record that is mostly market beta.

CVaR penalty. CVaR at $\alpha = 0.95$ is the expected shortfall of the R-multiple series on the loss side, taken positive. Coherent and tail-sensitive, it is preferred over VaR for fat-tailed trading returns.

SF — Safety Factor. The “last 8 S_RAW” are the eight most recent scoring rounds, carried in the agent's persisted state — not eight slices of a single submission. Each round contributes one EMA update, from a cold-start prior of $\text{EMA}_0 = 300$.

Design Principles

- Scale-free — all metrics in R-multiples or dimensionless ratios.
- Anti-overfit — DSR (Bailey & López de Prado, 2014) penalises statistical noise.
- Tail-aware — CVaR (Rockafellar & Uryasev, 2000) as explicit penalty.
- Anti-gaming — Volume Axiom is a hard gate: failure forces S_RAW = 0.

3.2 SISTEMA — $\text{EMA} \times \text{CF} \times \text{SF}$

SISTEMA is the time-compounding score that governs all protocol economics, bounded in $[0, 1000]$.

$$\text{SISTEMA}(t) = \text{EMA}(t) \times \text{CF}(t) \times \text{SF}(t)$$

Factor	Formula	Meaning
EMA	$\text{EMA}(t) = 0.85 \times \text{EMA}(t-1) + 0.15 \times S_{\text{RAW}} \times 10$	Memory of past performance
CF	$\min(\sqrt{(\text{trades}/250)}, 1)$	Confidence factor — full at 250 trades
SF	$\min(\text{mean}(\text{last } 8 \text{ } S_{\text{RAW}})/45, 1) \times (1 - \sigma_s/25)$	Safety — penalises inconsistency

Score Bands

Band	SISTEMA	Interpretation
EXPERIMENTAL	0–199	Insufficient track record
PROMISING	200–399	Emerging performance
VERIFIED	400–599	Consistent performer
ELITE	600–799	Top-tier
LEGENDARY	800–1000	Exceptional

Reproducibility. Every random stream used by the Pf Percentile is seeded deterministically, so an identical trade record always yields an identical score. A score committed on-chain can therefore be recomputed and verified by any party holding the same trades — the score is auditable, not merely asserted.

3.3 Empirical Validation

The YAAF score was tested against future returns across three venues with different data qualities.

Venue	Data Quality	n	IC (YAAF)	IC (Sharpe)	IC (WR)
GMX V2	high (real collateral)	60	0.4362 (p=0.0005)	0.1747 (p=0.18)	0.2139 (p=0.10)
Hyperliquid	medium (R estimated)	268	-0.0305 (p=0.62)	0.1144 (p=0.06)	0.0171 (p=0.78)
Darwinex	low (Risk Engine)	~80	0.1411 (p<1e-30)	—	—

The methodological finding: YAAF's predictive power is directly proportional to the quality of the R-multiple input. When the collateral is explicit (GMX V2), YAAF is 2.5x more predictive than Sharpe. When estimated, the edge vanishes. This validates the Volume Axiom as a fundamental gate, not a decorative constraint.

3.4 Data Quality Tiers

Every score in Yelden carries an explicit data quality tier.

Tier	Definition	Multiplier
high	Real collateral / R-multiple from source	1.00
medium	R-multiple estimated from notional	0.50
low	Returns smoothed by intermediary	0.25

```
C_allocated = min(C_kelly, C_stage) × quality_multiplier
```

04 — The AIAgentRegistry

The AIAgentRegistry is the on-chain enforcement layer for YAAF scores. Deployed on Polygon Mainnet.

Registry address: `0xC6Aef54A0ea6AbFcC9fe61154b9b357E0DDF113E`

verify: polygonscan.com/address/0xC6Aef54A0ea6AbFcC9fe61154b9b357E0DDF113E

4.1 registerAgent()

An agent registers by submitting its address and collateral stake in USDC. The registry stores: agent address, data source, data quality tier, registration timestamp, current SISTEMA score, and score history.

4.2 updateScore()

Scores are updated by addresses holding SCORER_ROLE. Each update writes S_RAW (current round), SISTEMA (compounded), Stage, and Timestamp.

4.3 slash()

The slash mechanism fires on either drawdown or capital loss, whichever produces the higher slash amount. This closes the “24% loophole” — an agent that stops trading at 24% DD to avoid a 25% threshold is still slashed on the capital loss dimension.

Drawdown	Capital Loss	Slash (stake)
< 10%	< 10%	0
10–25%	< 10%	$0.4 \times (DD - 0.10) \times \text{stake}$
24%	20%	capital loss trigger overrides
≥ 25%	any	full slash

4.4 isEligible() — Integration Contract

Any protocol, vault, or fund can integrate Yelden in two lines:

```
IAgentRegistry registry = IAgentRegistry(  
    0xC6Aef54A0ea6AbFcC9fe61154b9b357E0DDF113E  
);  
require(registry.isEligible(agent), "Agent not eligible");
```

isEligible() returns true when: SISTEMA ≥ 400 (VERIFIED or above), data quality tier is high or medium, and no pending slash event.

05 — Fee Model

Yelden operates entirely in USDC, USDT, and MATIC. No native token. No governance. No UBI.

5.1 Monthly Fee

$$\text{monthly_fee} = \text{base_rate} \times (1000 - \text{SISTEMA}) / 1000$$

At a base rate of 10 USDC/month:

- SISTEMA 1000 => 0 USDC (excellence is free)
- SISTEMA 500 => 5 USDC
- SISTEMA 0 => 10 USDC (self-eliminates)

Fees flow to the protocol treasury.

5.2 Slash Schedule

Level	Trigger	Penalty
WARNING	Grade 1 misconduct	10% of stake
SUSPENSION	Grade 2 misconduct	50% of stake
BAN	Critical fraud	100% of stake + deregistration

06 — Live Validation

6.1 Observatory — 478 Traders

Yelden operates a continuous observatory that scores autonomous traders across three venues:

Venue	Traders scored	Data quality
Hyperliquid	418	medium
GMX V2	60	high
Total	478	—

Scores update every 6 hours. The observatory is the empirical foundation for YAAF parameter calibration.

6.2 Live Trading Agents

Four autonomous agents run in production:

Agent	Venue	Status
Pepperstone bot	cTrader	Live — 16 closed trades
FTMO bot	cTrader	Live — 9 closed trades
OKX balancer	OKX	Live — rebalancing BTC/ETH/SOL/AVAX/gold/USD
IBKR equities	Interactive Brokers	Live — US stocks

All four will be registered in the AIAgentRegistry as they accumulate ≥ 30 closed trades, at which point YAAF produces statistically meaningful scores.

6.3 The First Validation — Markowitz Bot

The first agent to run under YAAF was the Markowitz Bot — a multi-strategy system executing across 7 sub-strategies with Mean-Variance Optimization. It logged 122 live trades with Sharpe 1.44 and max drawdown 6.81%. Its SISTEMA trajectory demonstrated that a real agent could be scored, tracked, and compared against alternatives. The Markowitz Bot has since been superseded by the cTrader agents above, which use the same scoring pipeline.

07 — Data Sources & Quality

Yelden integrates with venues that expose the data YAAF needs. The critical requirement is a real R-multiple — a PnL measured against the actual capital at risk.

Source	R-multiple basis	Quality
GMX V2	initialCollateralDeltaAmount (real)	high
cTrader API	Stop-loss and MAE (real)	high
Hyperliquid	Notional $\times$ 1% (estimated)	medium
dYdX	Notional $\times$ 1% (estimated)	medium
Darwinex Zero	Equity curve (smoothed)	low

08 — Roadmap

Phase	Timeline	Milestone
Foundation	Complete	YAAF v5 · AI-AgentRegistry on Polygon · Observatory (478 traders) · github.com/yeldenfund/yaaf
cTrader Integration	Q4 2026	cTrader API collector · Pepperstone + FTMO registered on-chain
Multi-Agent Onboarding	Q1 2027	10+ agents registered · first external agents
Genesis Scorer Program	Q1 2027	3–5 independent scorers · multi-scorer consensus
Formal Audit	Q2 2027	Code4rena or Sherlock audit
Chainlink Oracle	Q2 2027	Decentralised score submission

09 — Known Limitations

We document known limitations honestly. A protocol that cannot identify its own weaknesses cannot be trusted to improve.

- Single scorer. Currently the founder holds SCORER_ROLE. The Genesis Scorer Program is the fix (Q1 2027).
- No formal audit. Code is pre-audit. Code4rena / Sherlock engagement planned for Q2 2027.
- cTrader integration in progress. The Pepperstone and FTMO agents are live but not yet scored on-chain.
- Data quality variance. Scores from Hyperliquid and Darwinex carry explicit quality penalties.
- Short track record. The current agent fleet has 25 closed trades across two cTrader accounts.
- cTrader R-multiple via MAE. Maximum Adverse Excursion is used as a proxy for risk when stop-loss data is not explicitly available.

10 — Team

Paulo Longen — Founder

For 30 years, Paulo commanded systems where a wrong decision had real consequences — people, aircraft, missions. As a Colonel Engineer in the Brazilian Air Force, he spent his career designing and operating critical infrastructure where failure was not an option.

He retired in 2024 and entered the world of algorithmic trading. What he found was the opposite of everything he had spent three decades building: autonomous agents managing capital with no accountability layer at all. No skin in the game. No mechanism to distinguish skill from luck. No consequence for failure.

“I spent 30 years building systems that could not fail silently. Autonomous trading agents are the opposite: they fail silently, restart under a new address, and nobody notices. Yelden closes that gap.”

Yelden Protocol is the infrastructure he wished existed. Validated empirically ($\rho=0.44$, $p<0.001$ on GMX V2), deployed on Polygon mainnet, with four live trading agents running in production.

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11 — References

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12 — Disclaimer

This document is for informational purposes only. It does not constitute an offer or solicitation to purchase any security, investment, or financial instrument. Yelden Protocol is experimental smart contract infrastructure. Smart contracts may contain bugs. Past performance of any agent does not guarantee future results. Protocol revenue and incentives are not guaranteed. All forward-looking statements involve risk and uncertainty.

The AIAgentRegistry address cited in this document is deployed on Polygon Mainnet and has not been formally audited. Users interacting with it do so at their own risk.

Yelden Protocol — yelden.fund — October 2026